Project Design Phase-II
Data Flow Diagram & User Stories

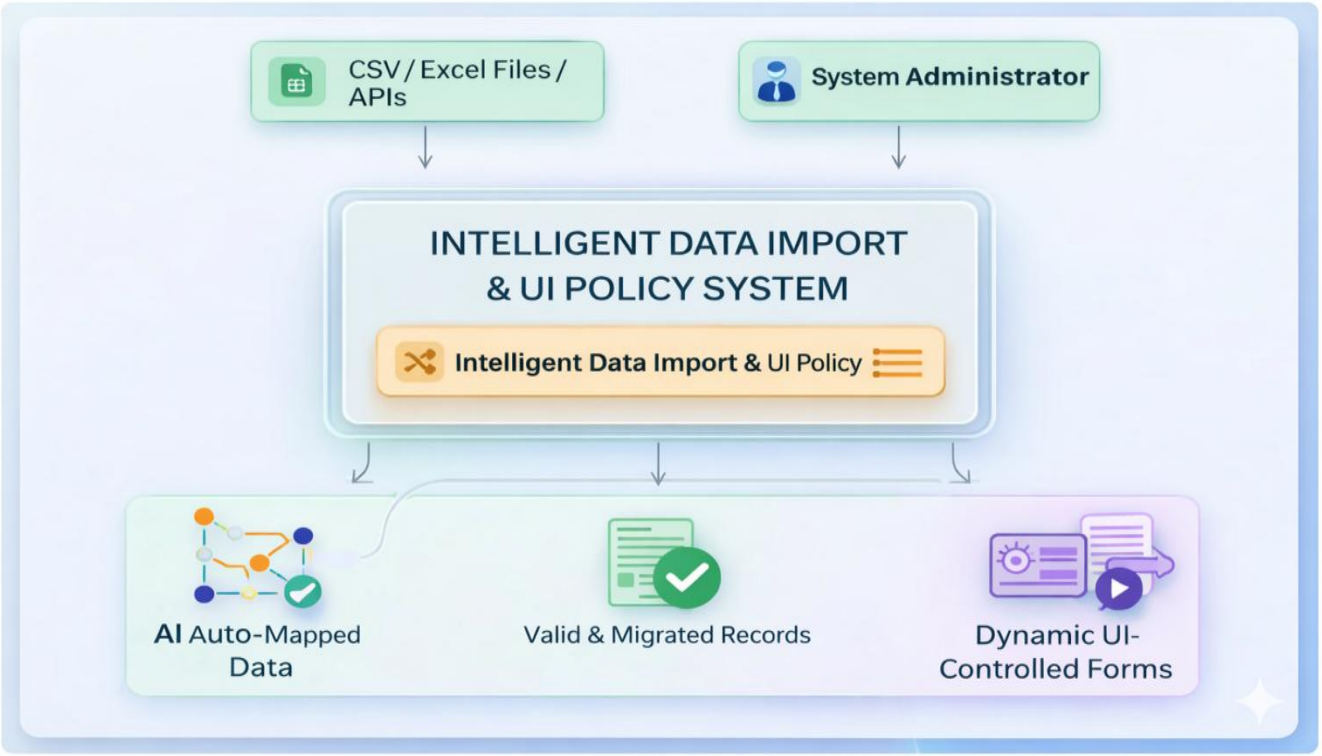
Date	08 April 2026
Team ID	SWTID-2026-8559
Project Name	Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets
Maximum Marks	4 Marks

Data Flow Diagrams:

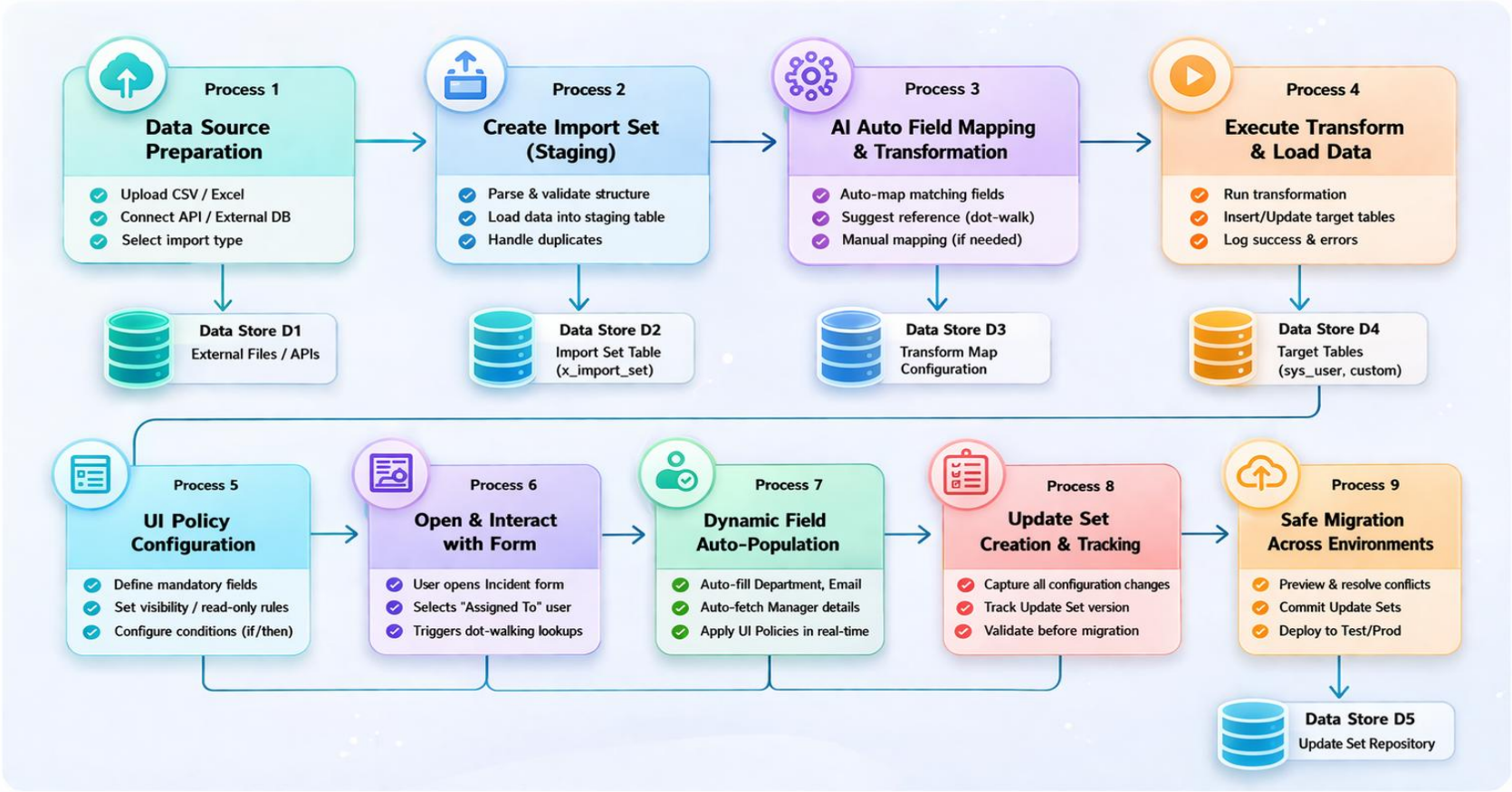
A **Data Flow Diagram (DFD)** in this project represents the flow of data within the **Intelligent UI Policy Enforcement and Data Import System in ServiceNow**. It visually explains how data is collected, processed, validated, transformed, and migrated across different environments while maintaining data integrity and automation

Example: (Simplified)

Level 0 DFD (Context Diagram)



Level 1 DFD (Detailed Process Flow)



Step	Process	Input	Output	Data Store
1	**Data Preparation & Source Integration**	CSV/Excel / API Data	Structured & validated dataset	D1: External Data Source / Local System
2	**Import Data into ServiceNow (Import Set)**	Structured dataset	Staging table records created	D2: Import Set Table
3	**AI-Based Field Mapping & Configuration**	Staging table data	Auto + manual mapping configuration	D3: Transform Map (AI Assisted)
4	**Data Validation & Integrity Check**	Mapped data	Cleaned & validated records, error logs	D4: Validation Logs / Audit Table
5	**Execute Data Transformation**	Validated data	Inserted/Updated records in target tables	D5: Target Tables (sys_user / custom tables)
6	**Dynamic UI Policy Enforcement**	User form interaction	Auto mandatory fields, visibility control, read-only fields	D6: UI Policy Engine
7	**Update Set Creation & Tracking**	Configuration changes	Version-controlled update sets	D7: Update Set Repository
8	**Safe Migration Across Environments**	Update sets	Error-free deployment to higher environments	D8: Migration Logs
9	**End User Interaction & Auto-Population**	User selection (e.g., Assigned To)	Auto-filled related fields (Manager, Email, Dept)	D5: Target Tables

User Stories:

Role: System Administrator

Story ID	User Story	Acceptance Criteria	Priority
US-ADMIN-1	As a System Administrator, I want to upload data from CSV/Excel/API sources so that I can import large datasets efficiently.	System allows file/API upload and validates format before processing.	High
US-ADMIN-2	As a System Administrator, I want the system to automatically create staging tables so that I can review imported data before processing.	Staging table is auto-created with preview and validation options.	High
US-ADMIN-3	As a System Administrator, I want AI-based auto field mapping so that manual mapping effort is reduced.	System suggests mappings automatically with high accuracy.	High
US-ADMIN-4	As a System Administrator, I want to manually adjust mappings using smart suggestions so that complex relationships are correctly handled.	System provides intelligent suggestions for reference fields and relationships.	High
US-ADMIN-5	As a System Administrator, I want real-time data validation and error detection so that incorrect data is prevented before transformation.	Validation engine detects duplicates, missing values, and incorrect formats.	High
US-ADMIN-6	As a System Administrator, I want to execute transformation with live status tracking so that I can monitor import success or failure.	Dashboard shows real-time status (Success, Failed, In Progress).	High
US-ADMIN-7	As a System Administrator, I want dynamic UI Policy enforcement so that fields become mandatory, visible, or read-only based on conditions.	UI fields change dynamically based on rules and user inputs.	High
US-ADMIN-8	As a System Administrator, I want to create and track Update Sets automatically so that all configuration changes are version controlled.	Update sets are auto-generated and tracked with history logs.	High

Story ID	User Story	Acceptance Criteria	Priority
US-ADMIN-9	As a System Administrator, I want safe migration across environments so that deployments are error-free.	System performs pre-check validation and conflict detection before migration.	High
US-ADMIN-10	As a System Administrator, I want detailed logs and audit trails so that I can debug and track system activities.	Logs include error details, timestamps, and user actions.	Medium

User Role 2: IT Support Agent

Story ID	User Story	Acceptance Criteria	Priority
US-AGENT-1	As an IT Support Agent, I want to create and manage incident tickets easily so that I can track and resolve user issues efficiently.	Agent can open, create, edit, and save incident records through a user-friendly interface.	High
US-AGENT-2	As an IT Support Agent, I want fields like Department, Email, and Manager to auto-populate when selecting a user so that manual lookup is eliminated.	Fields are automatically populated instantly using AI mapping and dot-walking without manual input.	High
US-AGENT-3	As an IT Support Agent, I want UI Policies to dynamically control form behavior so that only relevant fields are shown and required.	Fields become mandatory, visible, or read-only based on conditions in real-time.	High
US-AGENT-4	As an IT Support Agent, I want accurate and validated user data so that I can rely on the system for correct information.	Data shown in forms is validated, cleaned, and matches the latest imported dataset.	High
US-AGENT-5	As an IT Support Agent, I want reduced manual data entry through automation so that I can save time and increase productivity.	Auto-filled fields and smart suggestions reduce manual typing significantly.	High
US-AGENT-6	As an IT Support Agent, I want real-time feedback and error prevention while filling forms so that I avoid incorrect submissions.	System prevents submission of incomplete or invalid data using validation rules.	Medium

Story ID	User Story	Acceptance Criteria	Priority
US-AGENT-7	As an IT Support Agent, I want a smooth and fast user experience so that ticket creation and updates are efficient.	Forms load quickly, respond instantly, and require minimal user effort.	Medium

User Role 3: IT Manager

Story ID	User Story	Acceptance Criteria	Priority
US-MGR-1	As an IT Manager, I want all employee and system data to be consistent and accurate so that reporting and decision-making are reliable.	Data across target tables matches source data with validation ensuring zero discrepancies.	High
US-MGR-2	As an IT Manager, I want the data import process to be fully automated using AI so that manual effort and time consumption are reduced.	System supports scheduled/recurring imports with AI-based mapping and minimal manual intervention.	High
US-MGR-3	As an IT Manager, I want relationship mapping (User–Department–Manager) to be maintained automatically so that organizational hierarchy is always correct.	Reference fields are correctly mapped and maintained using intelligent transformation logic.	High
US-MGR-4	As an IT Manager, I want real-time dashboards and analytics so that I can monitor data import status and system performance.	Dashboard displays metrics like success rate, errors, processing time, and system health.	High
US-MGR-5	As an IT Manager, I want audit logs and tracking mechanisms so that I can monitor who performed actions and when.	Logs include user activity, timestamps, changes, and error tracking details.	Medium
US-MGR-6	As an IT Manager, I want safe and controlled migration across environments so that deployments are reliable and error-free.	System performs conflict detection, validation checks, and ensures successful migration.	High

User Role 4: HR Administrator

Story ID	User Story	Acceptance Criteria	Priority
US-HR-1	As an HR Administrator, I want to export employee data in a structured format so that it can be easily imported into the system.	Data is exported in standardized CSV/Excel/API format with required fields (Name, Email, Dept, Manager).	High
US-HR-2	As an HR Administrator, I want employee data updates to be automatically reflected in the system so that IT always has the latest information.	System triggers automatic data import and transformation when new data is available.	High
US-HR-3	As an HR Administrator, I want data validation mechanisms so that employee records maintain high quality and accuracy.	System validates email format, duplicate records, and correct relationships before processing.	High
US-HR-4	As an HR Administrator, I want minimal manual intervention during data handling so that errors are reduced.	AI-based mapping and validation reduce manual corrections significantly.	Medium
US-HR-5	As an HR Administrator, I want visibility into data import status so that I can confirm successful updates.	System provides status reports and notifications for data import and validation results.	Medium

Acceptance Criteria for Key User Stories

Story ID	Acceptance Criteria
US-ADMIN-1	✓ CSV/Excel/API data upload completes without errors✓ File format and structure are automatically validated✓ Data is successfully loaded into the staging (Import Set) table
US-ADMIN-3	✓ AI-based auto-mapping correctly maps matching fields✓ System provides intelligent suggestions for unmapped fields✓ Mapping accuracy reduces manual effort significantly
US-ADMIN-5	✓ Data validation detects duplicates, missing values, and format errors✓ Error logs are generated with clear descriptions✓ Only clean and validated data proceeds to transformation

Story ID	Acceptance Criteria
US-ADMIN-7	✓ UI Policies dynamically enforce mandatory, read-only, and visibility rules✓ Field behavior changes instantly based on conditions✓ Users cannot submit incomplete or invalid forms
US-ADMIN-9	✓ Update Sets are automatically created and tracked✓ Pre-migration validation detects conflicts✓ Migration completes successfully without data/configuration loss

Story ID	Acceptance Criteria
US-AGENT-2	✓ Selecting a user auto-populates Department, Email, and Manager fields✓ Data is fetched using AI mapping and dot-walking✓ Auto-filled fields are read-only and accurate
US-AGENT-3	✓ UI dynamically shows/hides fields based on conditions✓ Mandatory fields are enforced before submission✓ Form behavior adapts in real-time
US-AGENT-4	✓ Manager details are auto-populated and displayed as a reference link✓ Clicking the Manager field opens the related record✓ Relationship mapping is accurate and up-to-date
US-AGENT-6	✓ System prevents submission of incomplete or invalid data✓ Real-time validation messages are shown to users✓ Errors are clearly highlighted and guided for correction